CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL, COIMBATORE

CIRS

SAMPLE QUESTION PAPER

CLASS XI (CBSE)
PHYSICS

SECTION A

01. The time period of a simple pendulum is 2 sec. If its length is increased by four times, the time period becomes

a. 4 sec b. 6 sec c. 8 sec d. 2 sec

ANSWER	

02. In the absence of air resistance, if an object were to fall freely near the surface of the Moon:

a. Its velocity would never exceed 10m/s.
b. Its acceleration would gradually decrease.
c. The acceleration is constant.
d. It will fall with a constant speed.

ANSWER	

03. A real image is formed by a convex lens. If we put it in contact with a concave lens and the combination again forms a real image, which one of the following is true for the new image formed by this combination ?

a. shifts towards the lens system
b. shifts away from the lens system
c. remains at the original position
d. None of these

ANSWER	

04. Substances whose conductivity depends on ions are called

a. Electrolytes b. Conductors c. Insulators d. Semiconductors

ANSWER	

05. Heavy water in a nuclear reactor is used as

a. Moderator b. Protective shield c. Control rods d. Coolants

ANSWER	

06. The more readily fissionable isotope of uranium has an atomic mass of

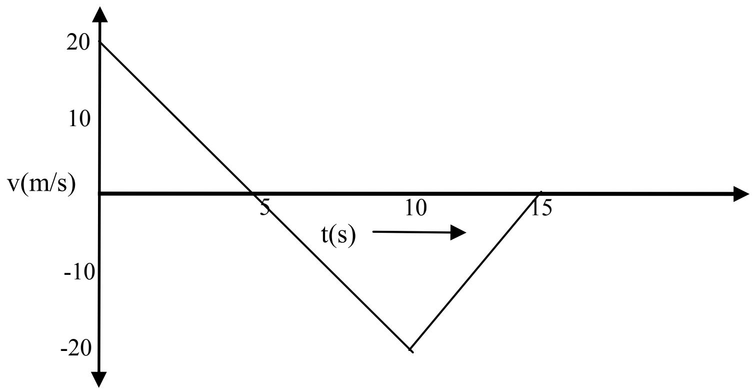
a. 238 b. 235 c. 236 d. 234

ANSWER	

SECTION B

07. A sprinter, starting from rest, reaches his full speed of 9 ms^{-1} in 1.5 s. What is his average acceleration?

08. For the following velocity-time graph, draw the corresponding acceleration-time graph.



09. Here is a picture of a parachutist, descending at a constant speed.

(a) What are the forces acting?
(b) What can you say about the magnitude of the forces?

10. An electrical appliance is rated as 2.5kW, 240V.
(a) What current does it need to draw in order to operate?
(b) How much energy would it consume in 2 hours?

11. Distinguish between Nuclear Fission and Nuclear Fusion. Give any two advantages and disadvantages of nuclear energy.

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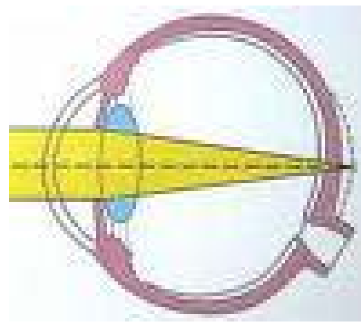
SAMPLE QUESTION PAPER

SECTION C

12. A ball is thrown downward with speed 15 m/s from the roof of a 30m building. At the same instant, another ball is thrown upward with speed 15m/s from ground level. When and where will these two balls meet each other?

13. The following figure shows a person suffering from an eye defect.



a. Name the defect.
b. Suggest the type of lens used to correct this defect.
c. Your friend has the same defect with near point at 120 cm. Calculate the power of lens used to correct this defect.

14. If a 15 V battery is connected to the arrangement of resistances given below, calculate

(i) the total effective resistance of the arrangement and
(ii) the total current flowing in the circuit when the key is closed.